

Scenario 1: Employee Salary Analysis

Question:

Find the names and salaries of employees who earn more than the average salary in the company.

Steps:

1. Calculate the average salary of all employees in the company using AVG() function.
2. Using WHERE, select the employees whose salary is greater than the average salary
3. Using SELECT, retrieve and display the names and salaries of these selected employees.

Scenario 2: Customer Orders without Matching Records

Question:

Retrieve a list of customer names who have not placed any orders.

Steps:

1. SELECT the names of customers from the Customer table.
2. Use a LEFT JOIN to connect the Customer table with the Order table, based on customer ID.
3. WHERE clause: Filter to find customers who do not have any matching records in the Order table (WHERE Order.customer_id IS NULL or WHERE NOT EXISTS a related order).
4. Retrieve and display the names of these customers.

Scenario 3: Product Sales Summary

Question:

Display the total sales amount for each product.

Steps:

1. SELECT the product identifier (such as product name or product_id) and sales amount from the sales/order details table.
2. Use the SUM aggregation function to calculate the total sales amount for each product.
3. Use GROUP BY on the product identifier to summarize sales for each product.

Scenario 4: Department-Wise Employee Count

Question:

List each department name with the number of employees working in it.

Steps:

1. SELECT the department name and employee ID (or any unique identifier) from the Employee table, possibly joining with the Department table if needed.
2. Use the COUNT aggregation function to count the number of employees in each department.
3. Use GROUP BY on the department name to group employees by each department.

Scenario 5: Top 3 Highest Sales

Question:

Find the top 3 highest sales transactions.

Steps:

1. SELECT the transaction details (such as transaction ID, sales amount, date, etc.) from the Sales/Transactions table.
2. Use the ORDER BY keyword to sort the transactions by sales amount in descending order (ORDER BY sales_amount DESC).
3. Use the LIMIT keyword to restrict the results to the top 3 records (LIMIT 3).

Scenario 6: Calculate Employee Salary Ranks by Department

Question:

Write a query to display each employee's name, department name, salary, and their salary rank within their respective department.

Steps:

1. SELECT the employee's name, department name, and salary from the Employee table, joining with the Department table if needed.
2. Use the RANK() window function to calculate each employee's salary rank within their department.
3. Perform an INNER JOIN to combine employee data with department information using department_id.
4. Display employee details with their salary rank using the ORDER BY clause.